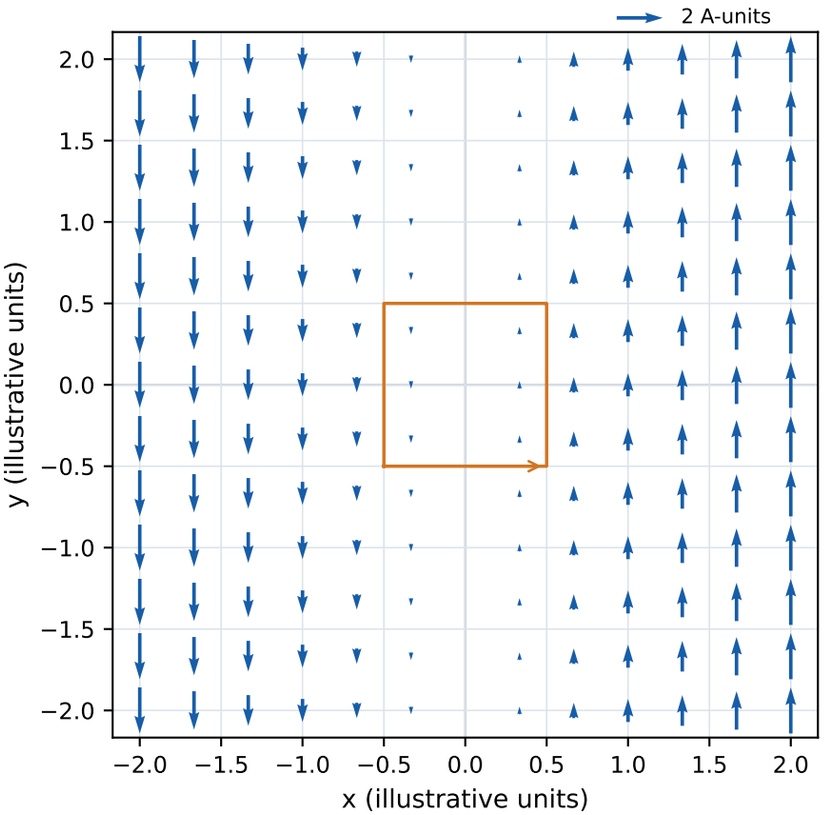
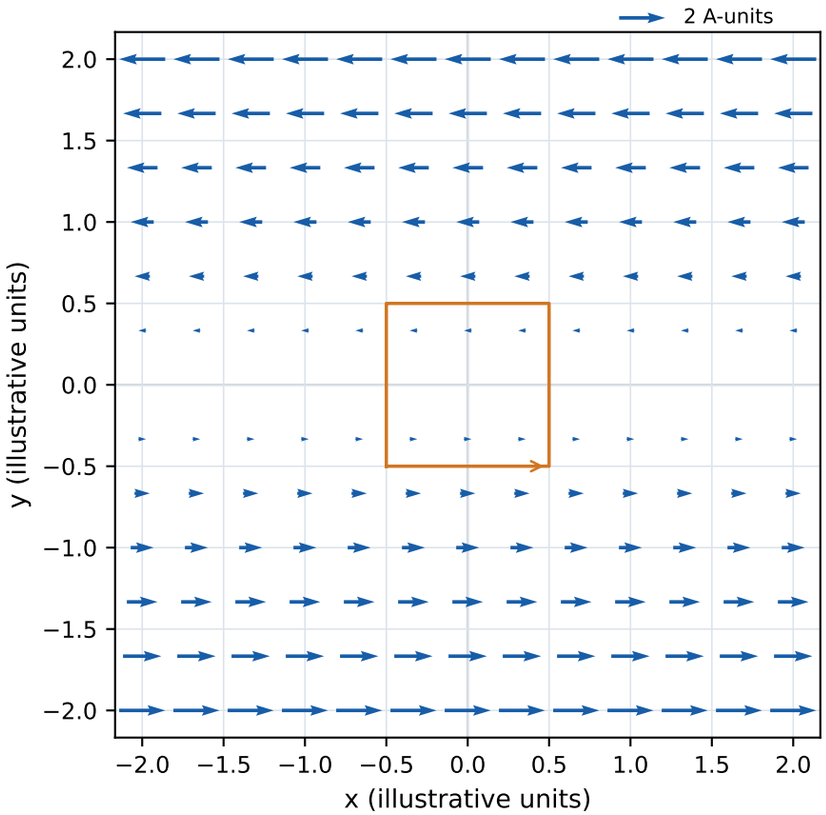
Exercise 3 · Different potentials, the same magnetic field

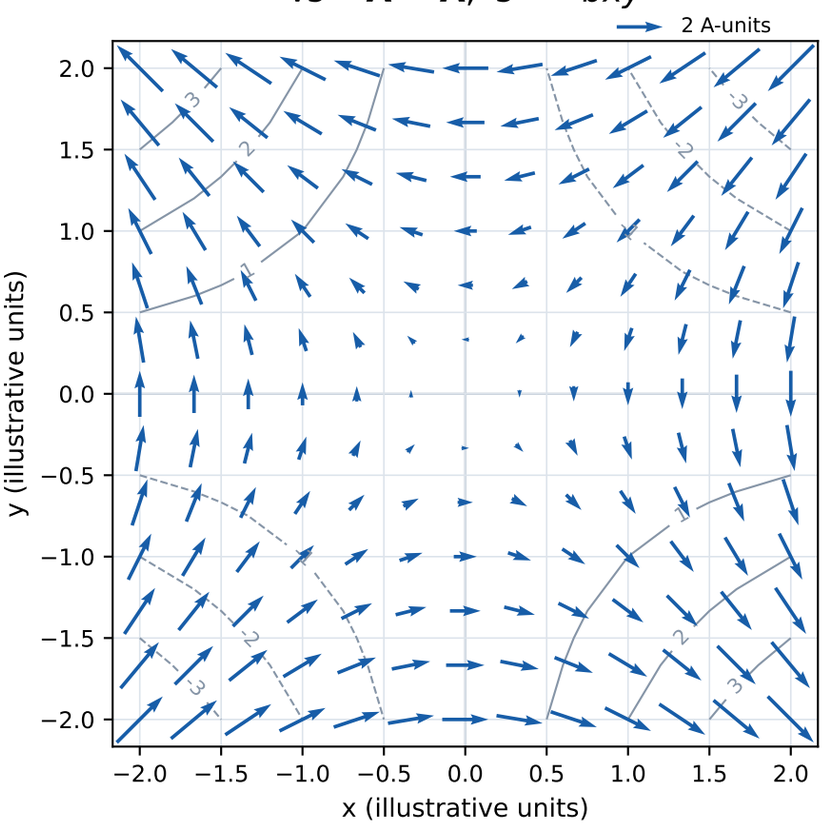
$\mathbf{A} = (0, bx, 0)$



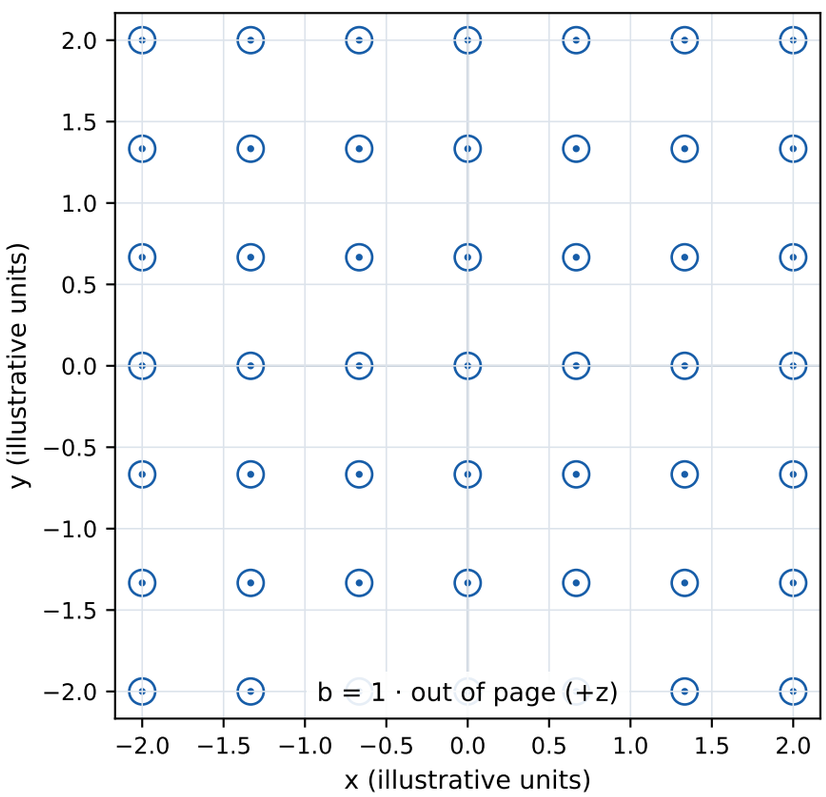
$\mathbf{A}' = (-by, 0, 0)$



$\nabla s = \mathbf{A}' - \mathbf{A}, \;\; s = -bxy$



$\mathbf{B} = \nabla \times \mathbf{A} = \nabla \times \mathbf{A}' = (0, 0, b)$



z = 0 · shared arrow scale for A, A′ and ∇s · orange squares: circulation loops